PML • CLASSROOM

K-MEANS CLUSTERING

From centroids to choosing K

A complete step-by-step lesson with one consistent
6-point example, calculations, graphs and classroom exercises.

LESSON MAP

What students will learn

The sequence follows how K-Means is actually used.

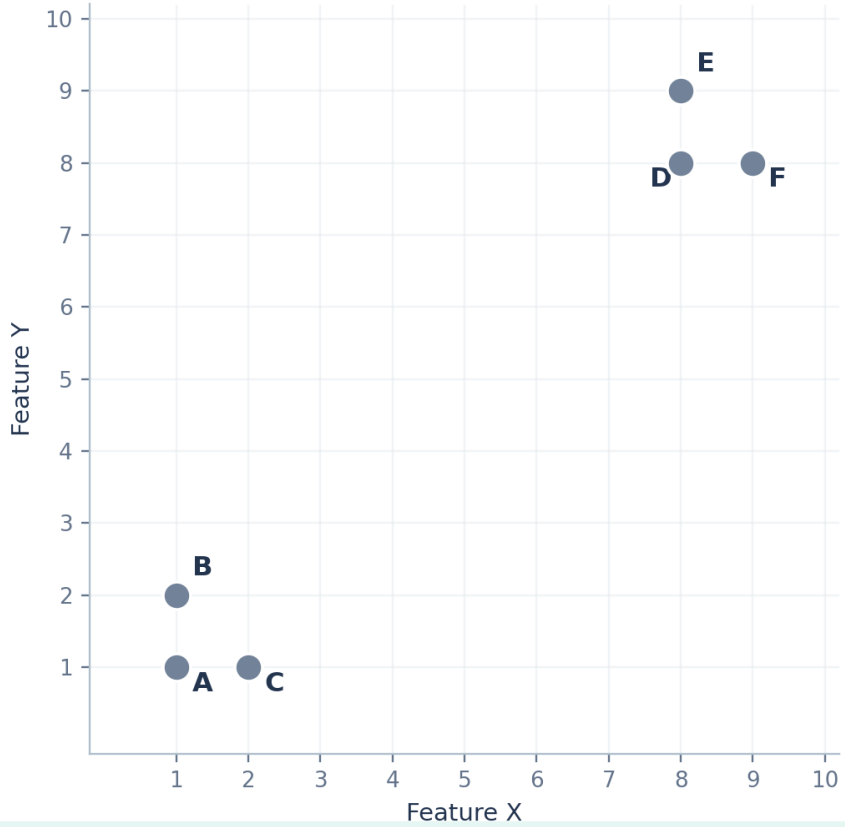
01	**Form the clusters**	Initialization → assignment → centroid update → convergence
02	**Measure compactness**	WCSS: squared distances to cluster centroids
03	**Choose K with Elbow**	Compare the reduction in WCSS as K increases
04	**Validate with Silhouette**	Compare cohesion and separation using a(i) and b(i)

WORKED EXAMPLE

Our six-point dataset

Two numerical features; cluster labels are initially unknown.

Point	X	Y
A	1	1
B	1	2
C	2	1
D	8	8
E	8	9
F	9	8



ASK STUDENTS What groups would you expect if K = 2?

Choose K and initialize centroids

We will deliberately choose simple starting centers for hand calculations.

NUMBER OF CLUSTERS

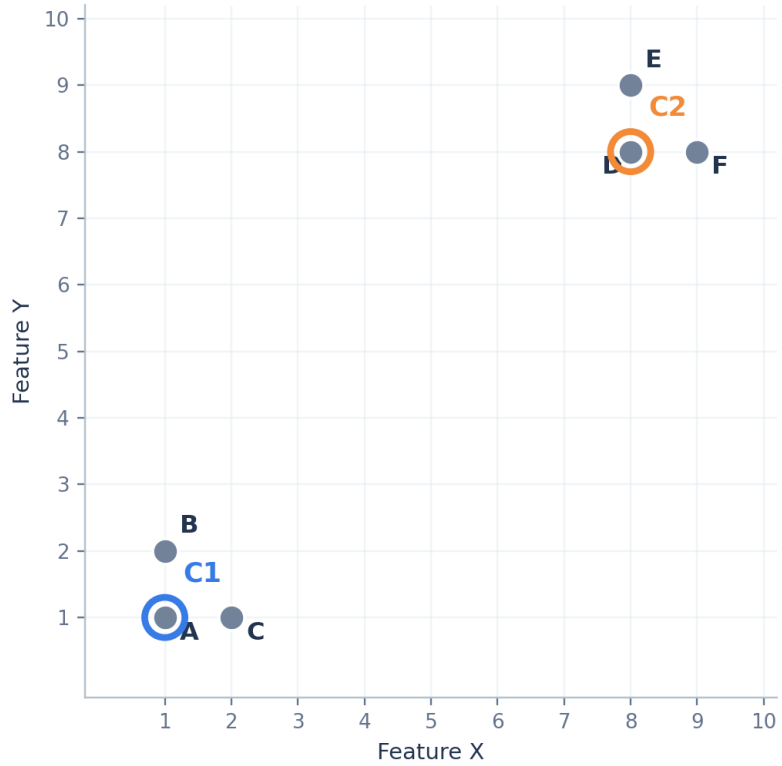
K = 2

INITIAL CENTROID 1

C1 = A = (1, 1)

INITIAL CENTROID 2

C2 = D = (8, 8)



ASK STUDENTS A centroid is a mean; must it remain an actual data point?

Why initialization matters

Different starting centers can lead to different local solutions.

Unhelpful starting choice

Both initial centers may fall in the same natural group. Example: A(1,1) and B(1,2).

K-Means++ approach

Choose first center randomly. Then favor points far from their nearest chosen center.

Assign each point to its nearest centroid

Use Euclidean distance; squared distance gives the same nearest center.

POINT B TO CENTROID 1

$$d(B,C1) = \sqrt{[(1-1)^2 + (2-1)^2]} = 1$$

POINT B TO CENTROID 2

$$d(B,C2) = \sqrt{[(1-8)^2 + (2-8)^2]} = 9.22$$

B is closer to C1, so B belongs to Cluster 1.

Point	d to C1	d to C2	Assign
A	0.00	9.90	1
B	1.00	9.22	1
C	1.00	9.22	1
D	9.90	0.00	2
E	10.63	1.00	2
F	10.63	1.00	2

Update each centroid using the mean

Centroids can move to positions where no data point exists.

CLUSTER 1: A, B, C

C1 = [(1+1+2)/3, (1+2+1)/3]

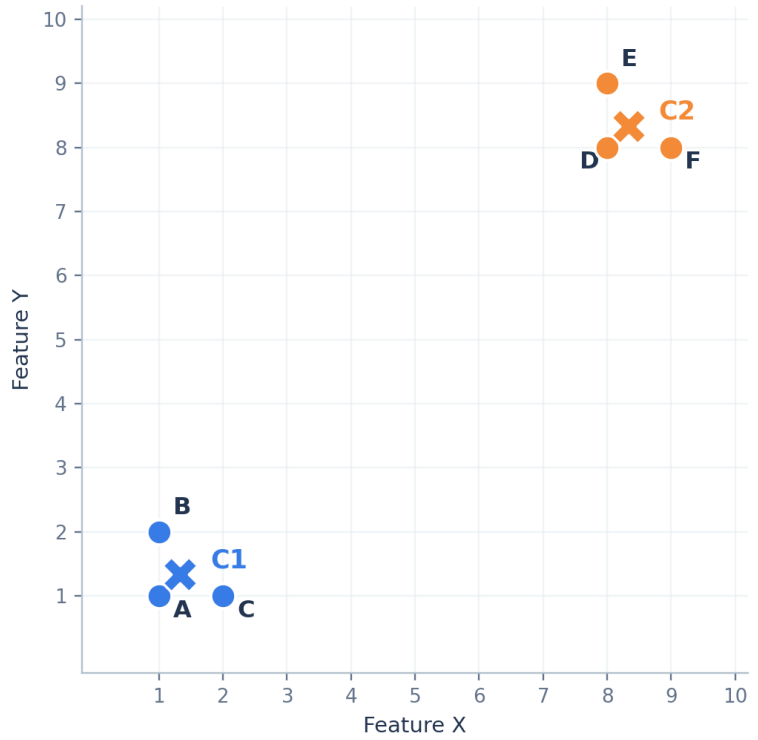
NEW CENTROID 1

C1 = (1.33, 1.33)

CLUSTER 2: D, E, F

C2 = [(8+8+9)/3, (8+9+8)/3]

New C2 = (8.33, 8.33)



Reassign, update and test convergence

Use the new centroids to calculate distances again.

Check	Cluster 1	Cluster 2
New centroids	(1.33, 1.33)	(8.33, 8.33)
Assigned points	A, B, C	D, E, F
After reassignment	A, B, C	D, E, F
After update	(1.33, 1.33)	(8.33, 8.33)

No points change clusters → centroids remain unchanged → CONVERGED.

The complete K-Means loop

The initialization stage ends; the ordinary iterative algorithm begins.

1. Select K and initial centers
2. Assign each point to nearest center
3. Recalculate each centroid as a mean
4. Repeat until stable or stopping criterion

CLUSTER QUALITY

What is K-Means trying to minimize?

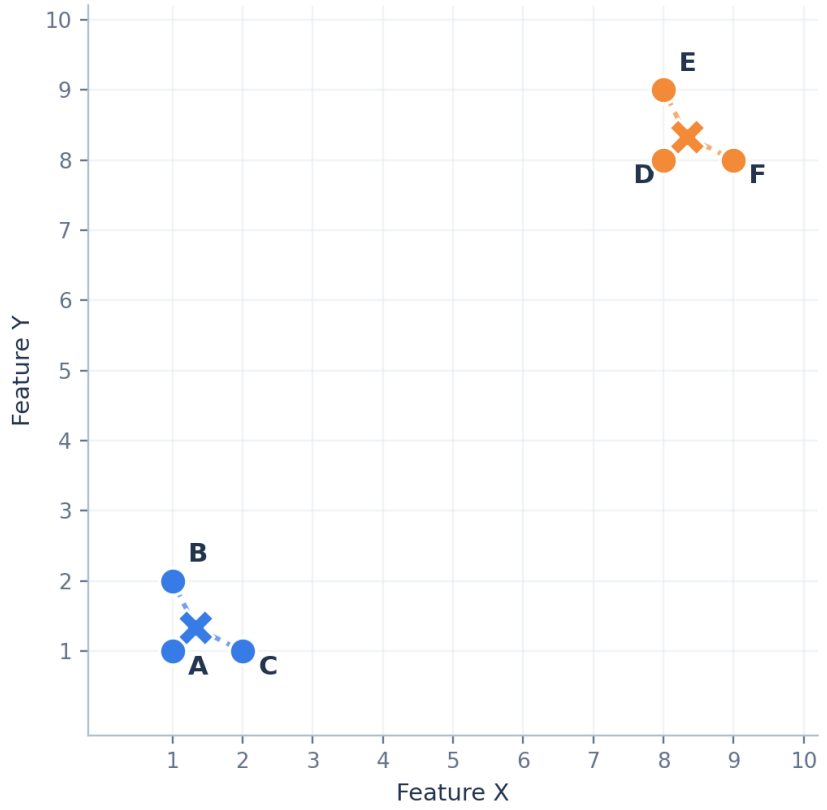
WCSS(Within-Cluster Sum of Squares) measures how tightly points gather around their own centroid.

OBJECTIVE

WCSS = Σ squared distance to assigned centroid

Interpretation

Lower WCSS means points are, overall, closer to the centers.



Calculate squared distance for Point A

Use the final Cluster 1 centroid C1 = (4/3, 4/3).

SQUARED EUCLIDEAN DISTANCE

$$d^2(A,C1) = (1 - 4/3)^2 + (1 - 4/3)^2$$

SIMPLIFY

$$= 1/9 + 1/9$$

RESULT

$$d^2(A,C1) = 2/9 = 0.2222$$

WCSS adds squared distances. It does not take the square root.

Calculate WCSS for both clusters

Using exact centroid means avoids unnecessary rounding error.

Cluster 1	Squared distance	Cluster 2	Squared distance
A	0.2222	D	0.2222
B	0.5556	E	0.5556
C	0.5556	F	0.5556
WCSS 1	**1.3333**	**WCSS 2**	**1.3333**

TOTAL INERTIA

WCSS = 1.3333 + 1.3333 = 2.6667

Why not choose the lowest WCSS?

With more clusters, WCSS usually becomes smaller, even if grouping is less useful.

K	Best WCSS	What it means
1	149.667	All six points share one centroid
2	**2.667**	**Two compact natural groups**
3	1.833	One group is split
4	1.000	Further fragmentation
5	0.500	Several singleton clusters
6	0.000	One point per cluster

Zero WCSS at K = 6 does not mean six useful segments.

ELBOW METHOD

Plot total WCSS against K

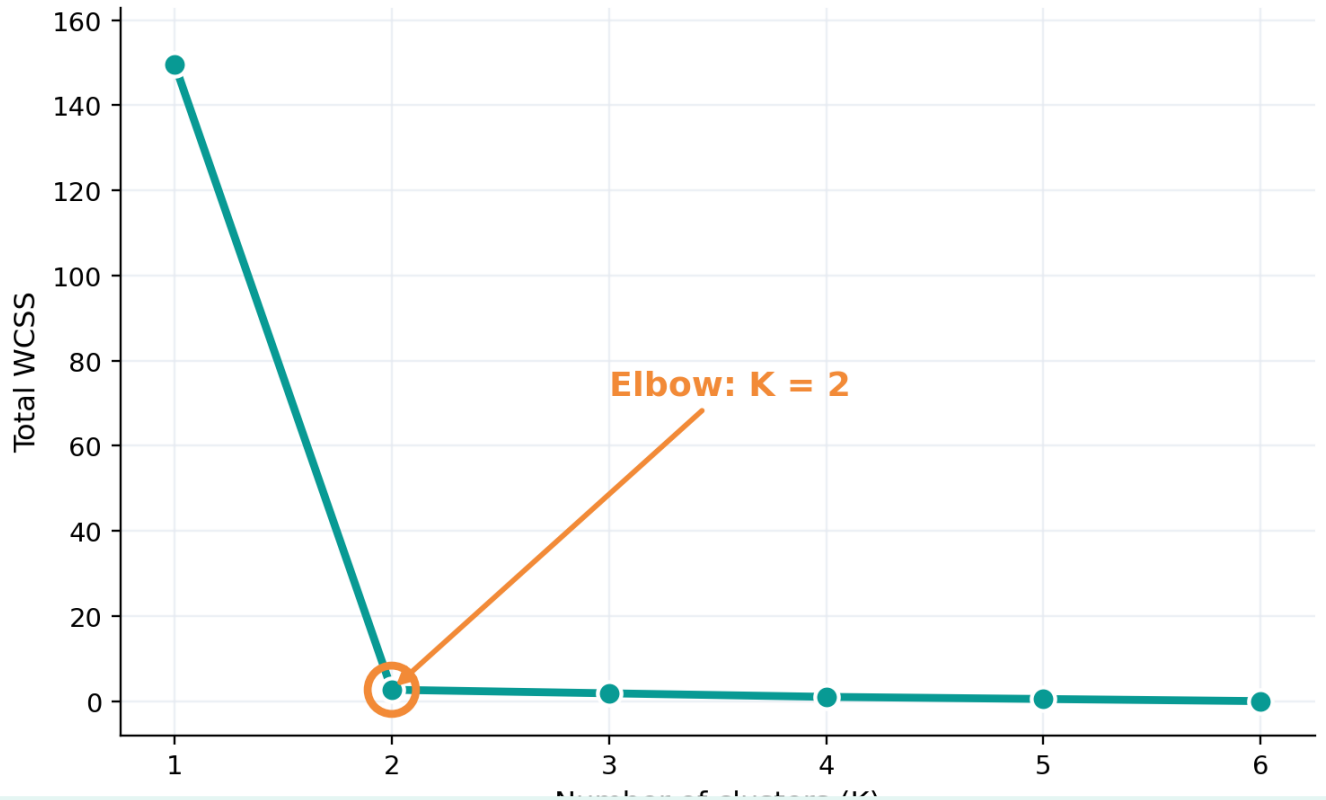
Choose a point where adding clusters gives diminishing returns.

Step 1

Run K-Means for several K values.

Step 2

Plot K vs total WCSS and look for a bend.



For our six points: the elbow is at K = 2.

Why the elbow is K = 2

Compare how much WCSS falls when an additional cluster is introduced.

Change in K	WCSS decrease
1 → 2	147.000
2 → 3	0.833
3 → 4	0.833
4 → 5	0.500
5 → 6	0.500

THE MAIN INSIGHT

1 → 2 is a huge improvement.

AFTER K = 2

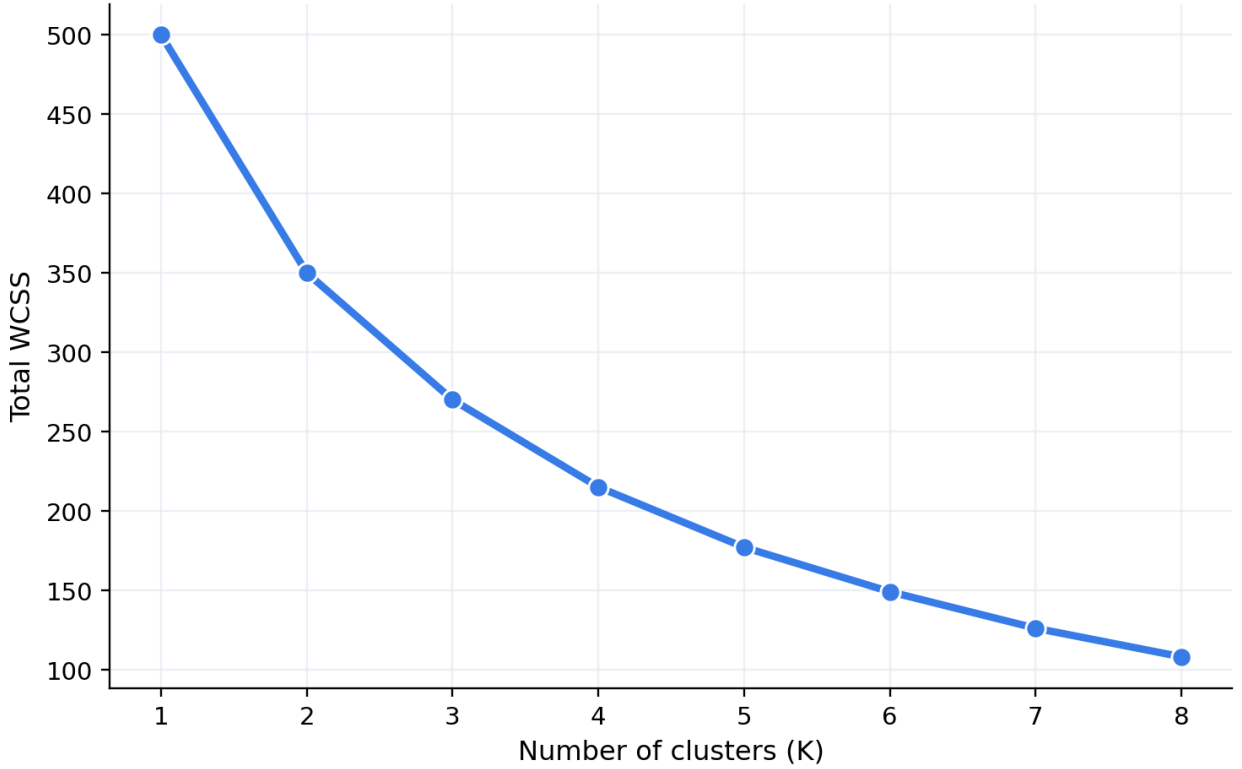
Only small gains remain.

Elbow is a heuristic, not a mathematical proof.

ELBOW / LIMITATION

What if there is no obvious elbow?

In real datasets WCSS may decline gradually instead of bending sharply.



SILHOUETTE / IDEA

Two questions for every data point

A good cluster is compact inside and separate from the other clusters.

a(i): cohesion

Average distance from point i to OTHER points in its OWN cluster.
Smaller is better.

b(i): separation

Smallest average distance from point i to points in ANOTHER cluster.
Larger is better.

Well-separated point → a(i) < b(i)

Calculate a(B): distance to own cluster

Point B = (1, 2); own cluster contains A, B and C.

DISTANCE B → A

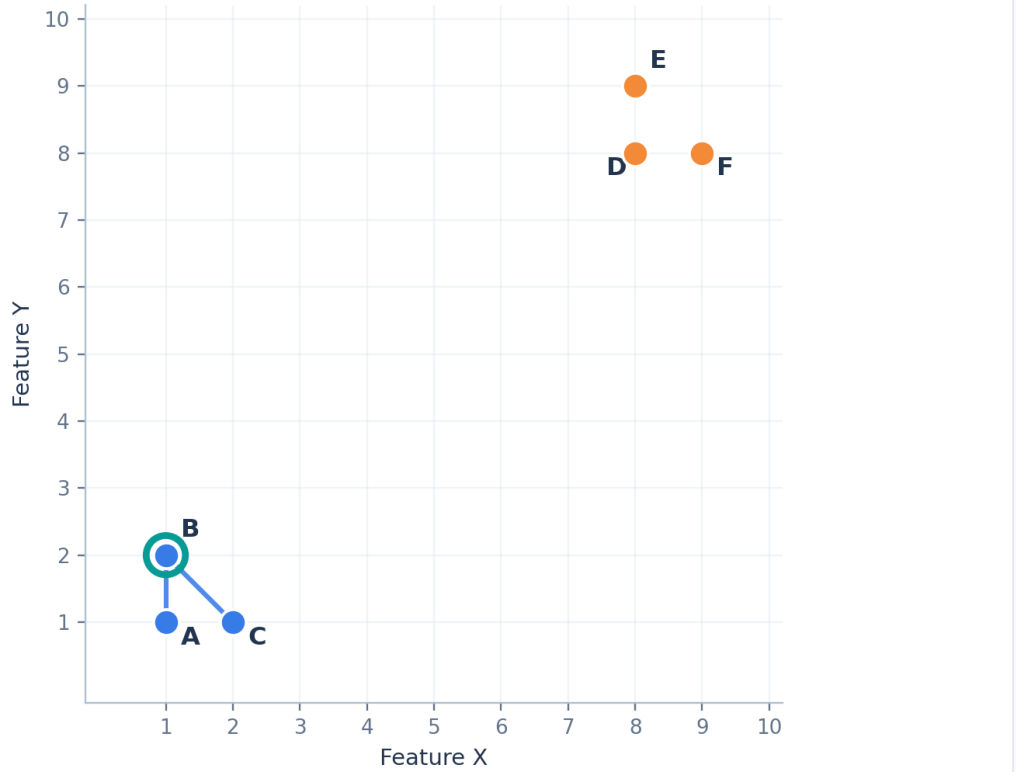
1.000

DISTANCE B → C

$\sqrt{2} = 1.414$

AVERAGE (EXCLUDE B ITSELF)

a(B) = (1 + 1.414)/2 = 1.207



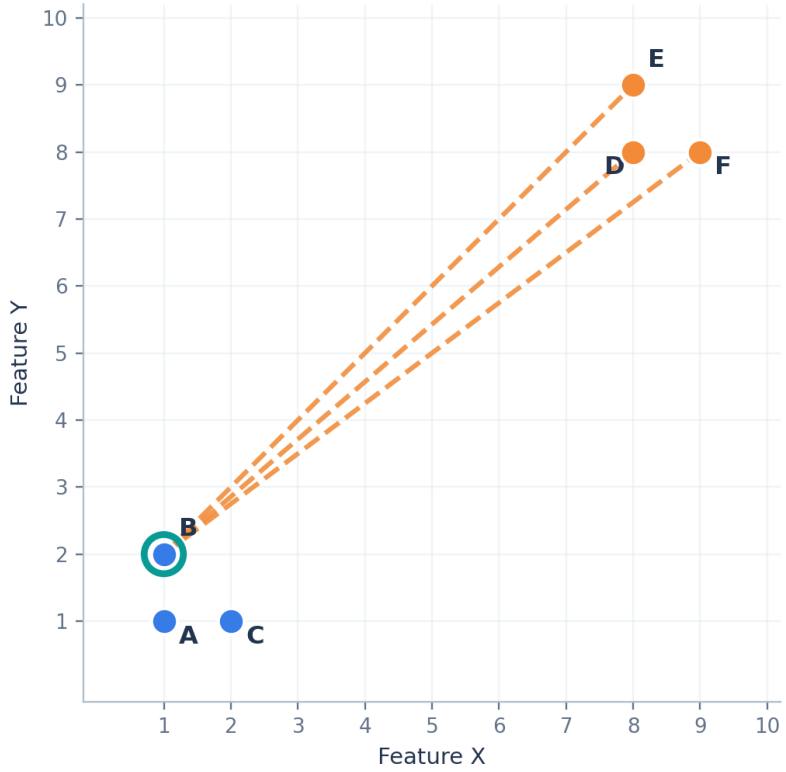
Calculate b(B): distance to the other cluster

For K = 2, there is only one alternative cluster: D, E and F.

Distance	Value
B → D	$\sqrt{85} = 9.220$
B → E	$\sqrt{98} = 9.899$
B → F	10.000

AVERAGE DISTANCE

b(B) = (9.220+9.899+10)/3 = 9.706



Calculate the Silhouette Score for Point B

Combine within-cluster compactness and distance to the nearest alternative cluster.

SILHOUETTE COEFFICIENT

$$s(i) = [b(i) - a(i)] / \max[a(i), b(i)]$$

a(B) — OWN CLUSTER

1.207

b(B) — OTHER CLUSTER

9.706

s(B) — RESULT

0.876

SUBSTITUTE

$$s(B) = (9.706 - 1.207) / 9.706 \approx 0.876$$

What does a Silhouette Score mean?

The coefficient ranges from −1 to +1.

Near +1

Point is much closer to its own cluster than the nearest other cluster.

Near 0

Average distance to its own cluster is similar to another cluster.

Negative

Another cluster is closer on average; assignment may be questionable.

EXAMPLE: NEGATIVE SCORE

$$a = 8,\ b = 3 \rightarrow s = (3 - 8)/8 = -0.625$$

A score close to 0 suggests overlap / boundary; it does not prove an exact geometric boundary.

SILHOUETTE / IMPORTANT DETAIL

What does b(i) mean if K > 2?

Choose the smallest mean distance across all OTHER clusters.

Cluster 1	Cluster 2	Cluster 3
Point i lives here	Mean distance = 4.5	Mean distance = 8.2

TAKE THE CLOSEST ALTERNATIVE CLUSTER

b(i) = min(4.5, 8.2) = 4.5

Do NOT average distances across all other clusters together.

From individual scores to one model score

Calculate one score per point, then take their mean.

Point	Cluster	a(i)	b(i)	s(i)
A	1	1.000	10.387	0.904
B	1	1.207	9.706	0.876
C	1	1.207	9.706	0.876
D	2	1.000	9.446	0.894
E	2	1.207	10.177	0.881
F	2	1.207	10.177	0.881

AVERAGE SILHOUETTE

(0.904 + 0.876 + 0.876 + 0.894 + 0.881 + 0.881)/6

RESULT

0.885

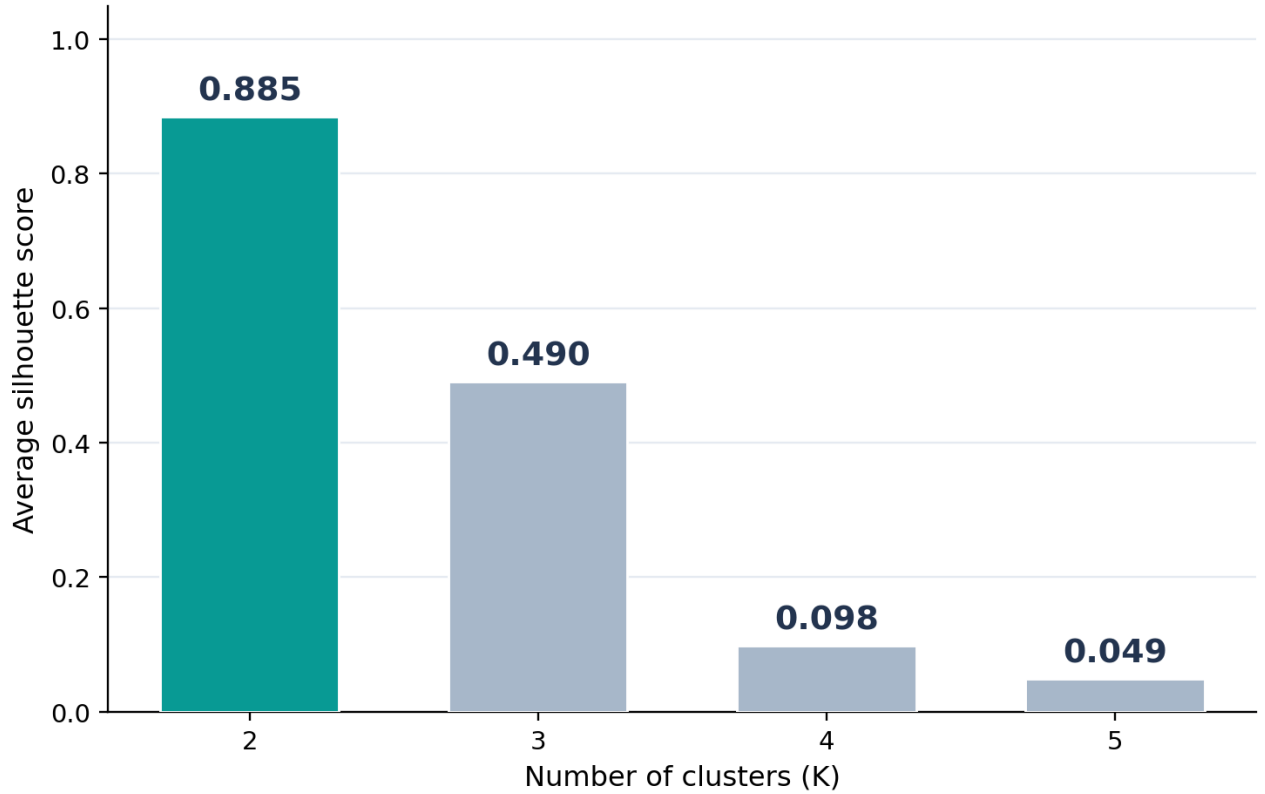
High: well-defined groups.

Compare the average score across K values

Higher mean Silhouette Score usually indicates better cohesion and separation.

K	Avg. score
2	0.885
3	0.490
4	0.098
5	0.049

Best of tested K: 2



Elbow vs Silhouette Score

Both help choose K, but they answer different questions.

Question	Elbow Method	Silhouette Score
What is measured?	Total WCSS / inertia	Cohesion and separation
Direction	Find a bend in decreasing WCSS	Higher average is better
Weakness	Elbow can be ambiguous	Can favor smaller/uneven clusters
Our six points	**Suggests K = 2**	**Highest score at K = 2**

GOOD PRACTICE

Use both — then inspect sizes, plots and business meaning.

PYTHON / IMPLEMENTATION

Fit K-Means and inspect its results

Clean scikit-learn code using the same six-point coordinates.

```
import numpy as np
from sklearn.cluster import KMeans

X = np.array([[1,1], [1,2], [2,1],
              [8,8], [8,9], [9,8]])

model = KMeans(n_clusters=2, init="k-means++",
               n_init=10, random_state=42)
labels = model.fit_predict(X)

print(model.cluster_centers_)
print(model.inertia_)
```

CLUSTER CENTERS

(1.33,1.33)

SECOND CENTER

(8.33,8.33)

WCSS / INERTIA

2.667

Run the Elbow and Silhouette comparisons

Keep the data and random seed consistent across candidate K values.

```
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score

wcss, scores = [], []
for k in range(1, 7):
    km = KMeans(n_clusters=k, n_init=10,
                random_state=42)
    labels = km.fit_predict(X)
    wcss.append(km.inertia_)
    if 2 <= k < len(X):
        scores.append((k, silhouette_score(X, labels)))

print('WCSS:', wcss)
print('Silhouette:', scores)
```

Elbow
K = 1...6

Silhouette
K = 2...5

K=1 and K=6: not defined.

Important practical considerations

Good mathematical clustering is not automatically useful clustering.

Scale features

Large-scale variables can dominate Euclidean distances.

Check outliers

K-Means and K-Means++ remain sensitive to extreme points.

Use multiple starts

Explicit n_init=10 explores different seeds; compare inertia.

Interpret segments

Check size, feature profiles, stability and domain meaning.

scikit-learn KMeans defaults to K-Means++ initialization; n_init is explicitly set here for reproducibility.

STUDENT EXERCISE

Check your understanding

Discuss first; reveal the answer slide only after students respond.

Q1 For B = (1,2), why is a(B) computed from A and C, not the centroid?

Q2 If a(i) = 6 and b(i) = 3, what is its Silhouette Score?

Q3 If WCSS keeps decreasing as K increases, why not choose K = n?

Q4 Which method would you use if the elbow graph has no clear bend?

TRAINER KEY

Suggested answers to the student exercise

Use for discussion or retain as a hidden teacher slide.

A1 Silhouette uses distances to other samples in the cluster; WCSS uses centroids.

A2 (3 − 6) / max(6,3) = −3/6 = −0.5. The point may fit better elsewhere.

A3 At K = n each point is its own cluster and WCSS = 0; segmentation is unhelpful.

A4 Compare candidate average Silhouette Scores and inspect cluster sizes/profiles.

FINAL RECAP

Five ideas to remember

Students should be able to explain each concept without looking at the slides.

1	**K-Means**	Assign → mean update → repeat
2	**K-Means++**	Better starting centers, not outlier protection
3	**WCSS**	Sum of squared distances to centroids
4	**Elbow**	Look for diminishing WCSS improvements
5	**Silhouette**	Compare average own-cluster and nearest-cluster distances

References: scikit-learn KMeans and silhouette_score documentation